



Internship: ALFIDO Tech Data Analytics Internship

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❖ Abstract

This project analyses sales and order data to evaluate overall business performance and identify opportunities to improve revenue and profitability. The analysis focuses on sales trends across regions, product categories, and time periods, to identify best- and worst-performing products and understand seasonal demand patterns.

Data was cleaned and prepared to ensure reliable analysis, after which key performance indicators (KPIs) such as revenue, profit margin, and average order value were calculated. Visual analytics were used to highlight performance trends, regional differences, and product-level contributions to sales and profit. These insights help pinpoint growth opportunities, inefficiencies, and potential risks within the current sales strategy.

Based on the findings, the project proposes targeted tactical recommendations to improve Alfido Tech's sales effectiveness, product positioning, and market focus. The analysis illustrates how structured sales analytics can support strategic planning, optimise decision-making, and drive sustainable business growth.

❖ Introduction

In a competitive business environment, understanding sales performance is essential for achieving sustainable growth and profitability. Organisations generate large volumes of sales and order data, but without proper analysis, this data remains underutilised and fails to support effective decision-making. Sales analytics enables businesses to evaluate revenue trends, identify profitable products, understand regional demand, and respond proactively to market changes.

This project focuses on analysing sales performance across different regions, product categories, and time periods to gain insights into business operations. By examining historical sales and order data, the study aims to identify the best- and worst-performing products, evaluate regional contributions to revenue, and analyse seasonal patterns that influence sales volume and profitability. Such analysis helps organizations optimize product offerings, pricing strategies, and regional focus.

The project applies systematic data cleaning, key performance indicator (KPI) computation, and exploratory data analysis techniques to transform raw sales data into meaningful insights. Visualisations are used to highlight trends, comparisons, and performance gaps in an interpretable manner. The insights derived from this analysis are intended to support strategic planning and guide tactical improvements for Alfido Tech's sales strategy, product positioning, and overall business performance.

❖ Data Description

The dataset used for this analysis is the **Superstore Sales** dataset, available at: <https://www.kaggle.com/datasets/bhanupratapbiswas/superstore-sales>

- **Number of rows:** 9,994 (approximate sales transactions)
- **Number of columns:** 20 distinct attributes related to orders and sales

Attributes Included

The dataset contains the following key fields:

- **Order Information:** Order ID, Order Date, Ship Date, Ship Mode
- **Customer Details:** Customer ID, Customer Name, Segment
- **Location:** Country, City, State, Region, Postal Code
- **Product Details:** Product ID, Category, Sub-Category, Product Name
- **Sales Metrics:** Sales, Quantity, Discount, Profit

Task 2

Dataset

<https://www.kaggle.com/datasets/bhanupratapbiswas/superstore-sales>

Goal

Analyze sales performance across regions, categories and time. Identify best/worst performing products and seasonality.

Requirements

- Clean and prepare sales & order data
- Create KPIs: revenue, profit margin, average order value, conversion metrics (if available)
- Build visualizations showing trends, top products, and regional performance
- Suggest 5 tactical improvements for Alfido Tech's sales or offerings

Objectives

- To analyze overall sales performance across regions, product categories, and time periods.
- To identify top-performing and underperforming products based on sales and profit.
- To examine seasonal trends and patterns affecting sales behavior.
- To derive actionable insights that support data-driven business decisions.

❖ Data Preprocessing

The dataset was imported using the pandas library with appropriate character encoding. The structure of the data, column data types, and completeness were verified using information and descriptive summary functions. Missing values were identified and removed to ensure data consistency and reliability. Records containing non-positive sales values were filtered out to eliminate invalid transactions. The order date variable was converted from string format to datetime format, after which new time-based features such as year, quarter, and month were extracted. Statistical summaries were generated for numerical and categorical variables to validate data distribution and consistency. After these steps, the dataset was cleaned and prepared for further analysis, visualization, and KPI evaluation.

❖ Methodology and Implementation

1. Importing Libraries and Customer Behavior Data

```
# Importing necessary libraries
```

```
import numpy as np
```

```
import pandas as pd
```

```
import datetime
```

```
df=pd.read_csv('/content/superstore_final_dataset (1).csv', encoding='latin')
```

```
df.head(10)
```

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Code	Region	Product_ID
0	1	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-BO10001796
1	2	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420.0	South	FUR-CH10000456
2	3	CA-2017-138688	12/6/2017	16/06/2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036.0	West	OFF-LA10000240

```
# Details of the data
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 9800 entries, 0 to 9799
```

```
Data columns (total 18 columns):
```

#	Column	Non-Null Count	Dtype
0	Row_ID	9800 non-null	int64
1	Order_ID	9800 non-null	object
2	Order_Date	9800 non-null	object
3	Ship_Date	9800 non-null	object
4	Ship_Mode	9800 non-null	object
5	Customer_ID	9800 non-null	object
6	Customer_Name	9800 non-null	object
7	Segment	9800 non-null	object
8	Country	9800 non-null	object
9	City	9800 non-null	object
10	State	9800 non-null	object
11	Postal_Code	9789 non-null	float64
12	Region	9800 non-null	object
13	Product_ID	9800 non-null	object
14	Category	9800 non-null	object
15	Sub_Category	9800 non-null	object
16	Product_Name	9800 non-null	object
17	Sales	9800 non-null	float64

```
dtypes: float64(2), int64(1), object(15)
```

```
memory usage: 1.3+ MB
```

2. EDA and Data Cleaning

```
df.isnull().sum()
```

Customer_Name	0
Segment	0
Country	0
City	0
State	0
Postal_Code	11
Region	0
Product_ID	0
Category	0
Sub_Category	0
Product_Name	0
Sales	0

```
df.dropna(inplace=True)
```

```
df.isnull().sum()
```

Customer_Name	0
Segment	0
Country	0
City	0
State	0
Postal_Code	0
Region	0
Product_ID	0
Category	0
Sub_Category	0
Product_Name	0
Sales	0

```
#Descriptive statistics
df.describe()
```

	Row_ID	Order_Date	Ship_Date	Postal_Code	Sales
count	9789.000000	9789	9789	9789.000000	9789.000000
mean	4896.705588	2017-05-01 02:51:57.866993664	2017-05-05 01:56:03.898253056	55273.322403	230.116193
min	1.000000	2015-01-03 00:00:00	2015-01-07 00:00:00	1040.000000	0.444000
25%	2449.000000	2016-05-23 00:00:00	2016-05-27 00:00:00	23223.000000	17.248000
50%	4896.000000	2017-06-26 00:00:00	2017-06-29 00:00:00	58103.000000	54.384000
75%	7344.000000	2018-05-15 00:00:00	2018-05-19 00:00:00	90008.000000	210.392000
max	9800.000000	2018-12-30 00:00:00	2019-01-05 00:00:00	99301.000000	22638.480000
std	2827.486899	NaN	NaN	32041.223413	625.302079

Conclusion: The analysis of the Superstore Sales dataset shows a positive growth trend in revenue with noticeable seasonal patterns. Sales are highly concentrated, with a few products and regions contributing most of the revenue, particularly the West and East regions. Most transactions are of low to moderate value, while a small number of high-value orders significantly impact total sales. Overall, the business demonstrates strong performance with clear opportunities to improve sales consistency by strengthening mid-performing products and underperforming regions.

```
# Qualitative characters
df.describe(include='O')
```

	Order_ID	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Region	Product_ID	Category	Sub_Category	Product_Name
count	9789	9789	9789	9789	9789	9789	9789	9789	9789	9789	9789	9789	9789
unique	4916	4	793	793	3	1	529	48	4	1860	3	17	1848
top	CA-2018-100111	Standard Class	WB-21850	William Brown	Consumer	United States	New York City	California	West	TEC-AC-10003832	Office Supplies	Binders	Staple envelope
freq	14	5849	35	35	5096	9789	891	1946	3140	18	5903	1492	47

```
df = df[df['Sales'] > 0]
```

```
#Average of Sales
average_sales=df['Sales'].mean()
print(average_sales)
```

```
230.1161929410563
```

Conclusion : The qualitative analysis shows that sales are dominated by the Consumer segment, Standard Class shipping, and the West region (especially California). Office Supplies, particularly Binders, are the most frequently sold products. The average sales value of about 230 per order indicates moderate transaction sizes, with overall performance driven by high order volume and key regions.

❖ Revenue

```
#Revenue
Revenue=df['Sales'].sum()
print('Revenue:',Revenue)
```

Revenue: 2252607.4127

Conclusion : The total revenue generated from the dataset is approximately 2.25 million, indicating strong overall sales performance during the study period. This level of revenue, combined with the observed variability in order values, suggests that the business benefits from both high order volumes and select high-value transactions. Leveraging top-performing products and regions can further enhance revenue growth.

```
df['order_Year']=df['Order_Date'].dt.year
df['order_quater']=df['Order_Date'].dt.quarter
df['Order_Month']=df['Order_Date'].dt.month

print("DataFrame with new time components:")
df.head()
```

DataFrame with new time components:

Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	...	Postal_Code	Region	Product_ID	
0	1	CA-2017-152156	2017-11-08	2017-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420.0	South	FUR-BO-10001798
1	2	CA-2017-152156	2017-11-08	2017-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	42420.0	South	FUR-CH-10000454
2	3	CA-2017-138688	2017-06-12	2017-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...	90036.0	West	OFF-LA-10000240
3	4	US-2016-108966	2016-10-11	2016-10-18	Standard Class	SO-20335	Sean O Donnel	Consumer	United States	Fort Lauderdale	...	33311.0	South	FUR-TA-10000577
4	5	US-2016-108966	2016-10-11	2016-10-18	Standard Class	SO-20335	Sean O Donnel	Consumer	United States	Fort Lauderdale	...	33311.0	South	OFF-ST-10000760

5 rows x 21 columns

```
yearly_sales = df.groupby('order_Year')['Sales'].sum()
print("\nYearly Sales:\n")
yearly_sales.head()
```

Yearly Sales:

Sales	
order_Year	
2015	479856.2081
2016	454315.9054
2017	597225.4900
2018	721209.8092

dtype: float64

Conclusion : Yearly sales show an overall increasing trend, with a small drop in 2016 followed by strong growth in 2017 and peak sales in 2018, indicating improved business performance over time.

```
quarterly_sales = df.groupby(['order_Year', 'order_quater'])['Sales'].sum()
print("\nQuarterly Sales:\n")
quarterly_sales.head()
```

Quarterly Sales:

		Sales
order_Year	order_quater	
2015	1	73931.3960
	2	85874.0936
	3	142522.6063
	4	177528.1122
2016	1	62357.6870

dtype: float64

Conclusion : Quarterly sales show a clear seasonal pattern, with sales generally increasing from Q1 to Q4. The fourth quarter consistently records the highest sales, indicating strong year-end demand and the importance of Q4 in overall revenue generation.

```
Monthly_sales=df.groupby(['order_Year', 'Order_Month'])['Sales'].sum()
print('\nMonthly_sales\n')
display(Monthly_sales.head())
```

Monthly_sales

		Sales
order_Year	Order_Month	
2015	1	14205.707
	2	4519.892
	3	55205.797
	4	27906.855
	5	23644.303

dtype: float64

Conclusion : Monthly sales show strong fluctuations throughout the year, indicating clear seasonality. Sales generally start lower in early months and increase in later months, contributing to higher quarterly and yearly totals, especially toward the end of the year.

❖ Profit Margin

1

Profit margin cannot be calculated because the dataset does not include cost or profit information. Without these attributes, profitability cannot be derived from sales data alone, so profit-based KPIs

were excluded to avoid misleading results.

❖ Average order value (AOV)

```
aov=df.groupby('Order_ID')['Sales'].sum().mean()
print('AOV:',aov)
```

AOV: 458.2195713384866

Conclusion : The Average Order Value (AOV) is approximately 458, indicating that while many individual items have moderate prices, customers often purchase multiple items per order, resulting in higher overall order value. This suggests good cross-selling or bulk purchasing behavior.

❖ conversion metrics

Conversion metrics cannot be calculated because the dataset lacks traffic or lead-level data required to define conversions.

❖ KPIS(Key performance Indicators)

```
kpis = {
    "Revenue": Revenue,
    "Profit Margin (%)": "Cannot be calculated due to missing profit/cost data.",
    "Average Order Value": aov,
    "Conversion Metrics": "Cannot be calculated due to missing traffic/lead data."
}
print("KPI SUMMARY")
for k, v in kpis.items():
    print(f"{k}: {v}")
```

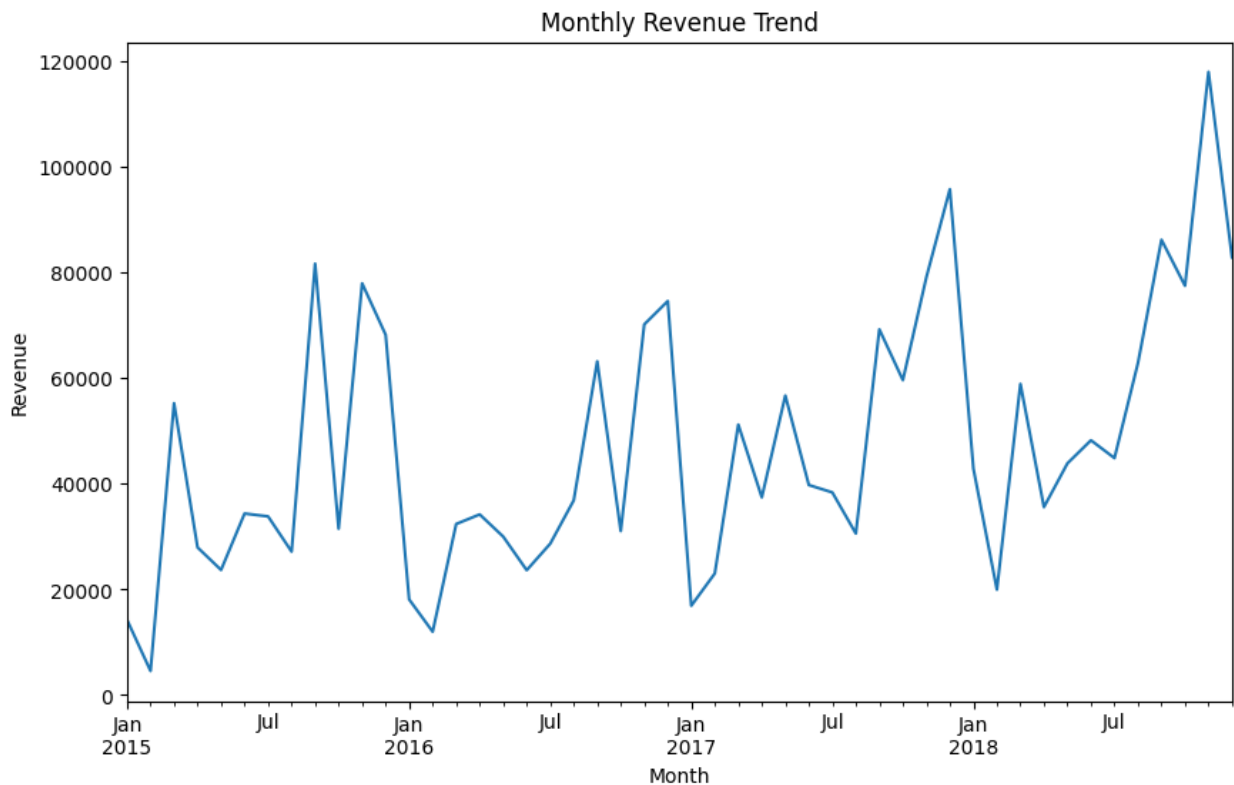
KPI SUMMARY
Revenue: 2252607.4127
Profit Margin (%): Cannot be calculated due to missing profit/cost data.
Average Order Value: 458.2195713384866
Conversion Metrics: Cannot be calculated due to missing traffic/lead data.

❖ Visualization

Revenue Trend Over Time

```
import matplotlib.pyplot as plt
# Revenue Trend Over Time
df['Month'] = df['Order_Date'].dt.to_period('M')
monthly_sales = df.groupby('Month')['Sales'].sum()

plt.figure(figsize=(10,6))
monthly_sales.plot()
plt.title("Monthly Revenue Trend")
plt.xlabel("Month")
plt.ylabel("Revenue")
plt.show()
```



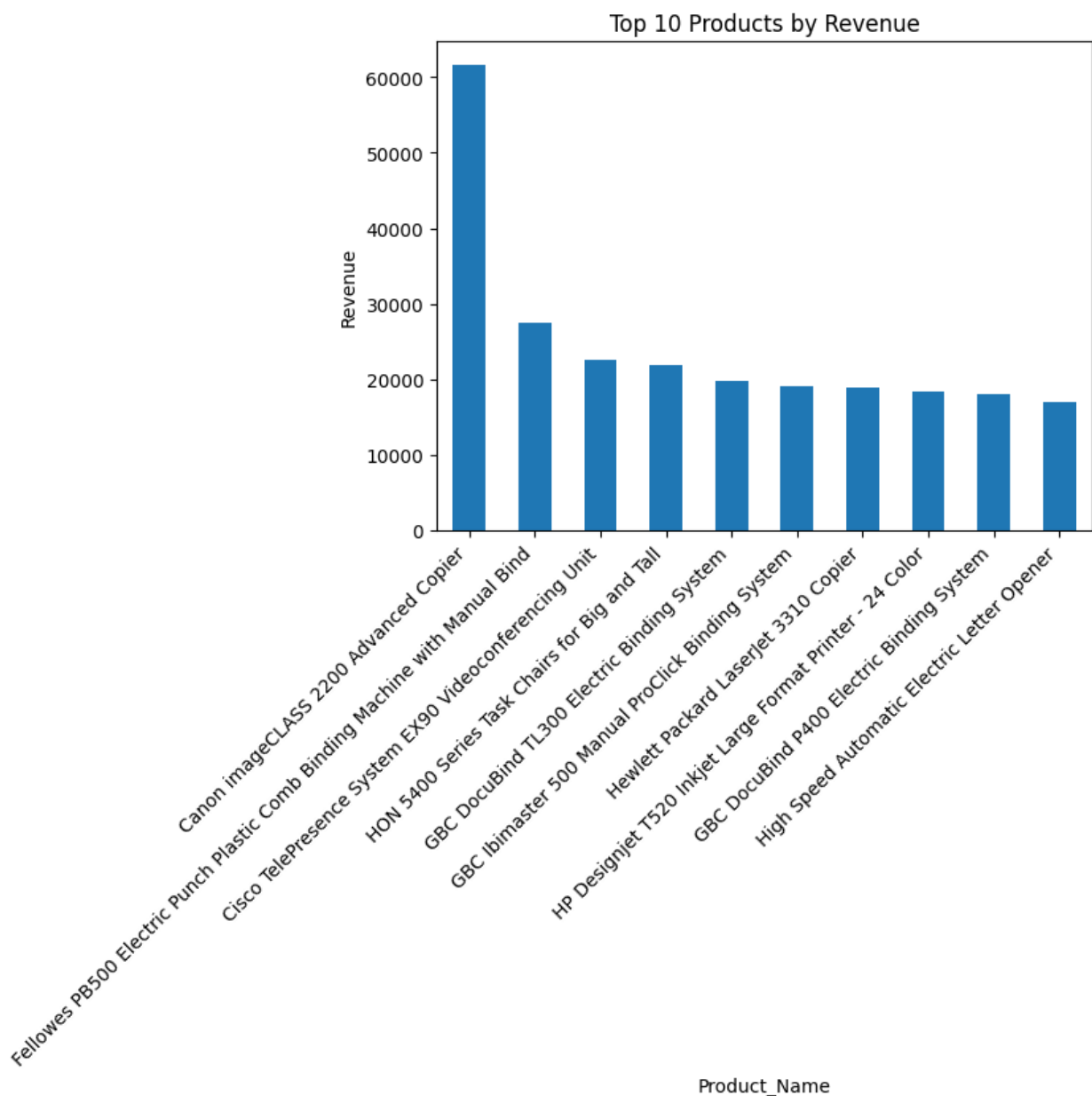
Conclusion : The monthly revenue trend shows a clear upward growth pattern over time, indicating that overall sales performance is improving year by year. While revenue fluctuates significantly from month to month, the peaks become higher in later years, suggesting successful expansion and increasing market demand.

❖ Top products by Revenue

```

top_products=df.groupby('Product_Name')['Sales'].sum().sort_values(ascending=False).head(10)
plt.figure()
top_products.plot(kind='bar')
plt.title("Top 10 Products by Revenue")
plt.ylabel("Revenue")
plt.xticks(rotation=45, ha='right')
plt.show()

```



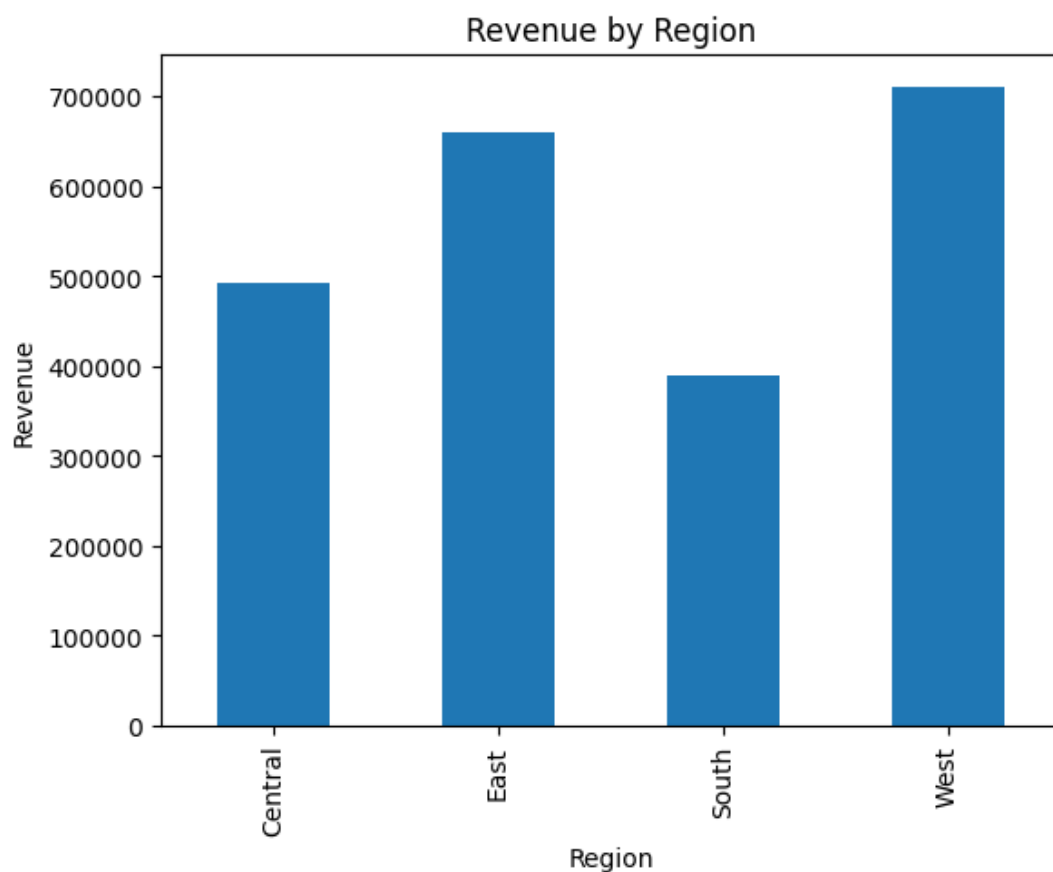
Conclusion : The bar chart highlights a¹ highly skewed revenue distribution, where a small number of products contribute a large share of total sales. This

indicates strong product concentration within the portfolio.

❖ Regional Performance

```
regional_sales = df.groupby('Region')['Sales'].sum()
```

```
plt.figure()  
regional_sales.plot(kind='bar')  
plt.title("Revenue by Region")  
plt.ylabel("Revenue")  
plt.show()
```



Conclusion: The regional revenue analysis reveals¹ significant variation in sales performance across regions, indicating uneven market penetration and growth opportunities.

❖ **Business Insights:**

1. **Revenue growth is improving over time**

Yearly and monthly analyses show a clear upward trend in sales, with strong performance in recent years, indicating successful business expansion and increasing customer demand.

2. **Sales are highly seasonal**

Quarterly and monthly patterns reveal that Q4 consistently generates the highest revenue, highlighting the importance of year-end promotions, festive sales, and inventory readiness during this period.

3. **Revenue is driven by few high-value orders**

The sales distribution is right-skewed, where a small number of large transactions significantly impact total revenue. Retaining high-value customers is critical for sustained growth.

4. **Strong regional concentration**

The West and East regions outperform others, while the South region lags behind. This indicates untapped market potential in underperforming regions.

5. **Product portfolio imbalance**

A limited number of products contribute a large share of revenue, increasing dependency risk. Mid-performing products offer opportunities for targeted promotion and bundling.

6. **Healthy Average Order Value (AOV)**

An AOV of around 458 suggests effective cross-selling and multi-item purchases, which can be further enhanced through personalized recommendations.

7. **Consumer segment dominance**

The Consumer segment contributes the highest number of orders, suggesting that customer-focused pricing and loyalty strategies can significantly impact sales.

8. **Standard shipping preference**

Most customers prefer Standard Class shipping, indicating cost sensitivity and opportunities to optimize logistics costs.

